

AAKRITI KUMAR

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CURRENT APPOINTMENT

Postdoctoral Researcher

Aug 2024 - Present

Kellogg School of Management, Northwestern University
Northwestern Institute on Complex Systems
Ryan Institute on Complexity

EDUCATION

PhD in Cognitive Science

2018 - 2024

University of California, Irvine
Thesis: Human Mental Models of Self, Others, and AI Agents

MS in Statistics

2018 - 2022

University of California, Irvine

BTech in Engineering

2011 - 2015

Indian Institute of Technology, Madras

RESEARCH EXPERTISE

Human-AI Collaboration, Behavioral Science, Cognitive Science, Statistical Modeling, Team Dynamics, Digital Experimentation, Computational Social Science

PEER-REVIEWED PUBLICATIONS

- [1] **A. Kumar**, N. Poungpeth, D. Yang, E. Farrell, B. Lambert, M. Groh (2026). [When Large Language Models are Reliable for Judging Empathic Communication](#). *Nature Machine Intelligence*
- [2] **A. Kumar**, R. Tham, M. Steyvers (2025). [Seeing Things Differently: The Role of Differing Perspectives in Advice-Taking](#). *Cognitive Science (Vol. 47)*
- [3] M. Steyvers, H. Tejada, **A. Kumar**, S. Karny, X. Hu, L. Mayer, P. Smyth (2025). [What large language models know and what people think they know](#). *Nature Machine Intelligence*
- [4] N. Kamali, K. Nakamura, **A. Kumar**, A. Chatzimpampas, J. Hullman, M. Groh (2025). [Characterizing Photorealism and Artifacts in Diffusion Model-Generated Images](#). *CHI 2025*
- [5] **A. Kumar**, P. Smyth, M. Steyvers (2023). [Differentiating mental models of self and others: A hierarchical framework for knowledge assessment](#). *Psychological Review*
- [6] M. Steyvers & **A. Kumar** (2023). [Three Challenges for AI-Assisted Decision-Making](#). *Perspectives on Psychological Science*
- [7] M. Kelly, **A. Kumar**, P. Smyth, M. Steyvers (2023). [Capturing Humans' Mental Models of AI: An Item Response Theory Approach](#). *FAccT 2023*
- [8] **A. Kumar**, K. Akash, S. Mehrotra, T. Misu, M. Steyvers (2023). [When Do Drivers Intervene In Autonomous Driving?](#) *Human Robot Interaction (Late-Breaking Report) 2023*
- [9] H. Lemus, **A. Kumar**, M. Steyvers (2022). [How Displaying AI Confidence Affects Reliance and Hybrid Human-AI Performance](#). *Proceedings of the Second International Conference on Hybrid Human-Machine Intelligence*.
- [10] H. Lemus, **A. Kumar**, P. Smyth, M. Steyvers (2022). [AI-assisted Decision-Making: A Cognitive Modeling Approach to Infer Latent Reliance Strategies](#). *Computational Brain & Behavior*

- [11] H. Lemus, **A. Kumar**, M. Steyvers (2022). [An Empirical Investigation of Reliance on AI-Assistance in a Noisy-Image Classification Task](#). *Proceedings of the First International Conference on Hybrid Human-Machine Intelligence*.
- [12] S. Chatterjee^{*1}, **A. Kumar***, P. Shenoy (2022). [Meta-Learning of Dynamic Policy Adjustments in Inhibitory Control Tasks](#). *Cognitive Science (Vol. 44)*
- [13] **A. Kumar**, T. Patel, A. Benjamin, M. Steyvers (2021). [Explaining Algorithm Aversion with Metacognitive Bandits](#). *Cognitive Science (Vol. 43)*
- [14] **A. Kumar**, A. Benjamin, A. Heathcote, M. Steyvers (2021). [Comparing models of learning and relearning in large-scale cognitive training data sets](#). *NPJ Science of Learning*.
- [15] J. Rouder, **A. Kumar**, J. Haaf (2019). [Why Most Studies of Individual Differences With Inhibition Tasks Are Bound To Fail](#). *Psych Bulletin & Review*

PREPRINTS & DRAFTS

- [P1] **A. Kumar**, N. Pongpeth, D. Yang, B. Lambert, M. Groh. [Practicing with Language Models Cultivates Human Empathic Communication](#). *arXiv*
- [P2] N. Dundas*, **A. Kumar***, Z. Guo, O. Badri, A. Koochek, M. Groh. Mapping Diagnostic Blind spots by Physicians and AI in Dermatology.
- [P3] N. Kamali, **A. Kumar**, A. Chatzimparmpas, C. Gerstner, J. Hullman, M. Groh. Guiding Human Visual Attention to Spot AI-Generated Images.
- [P4] **A. Kumar**, P. Petrides, K. Chauncey. Simulating AV System Delays to Study User Stress & Behavioral Intention.

RESEARCH IN PROGRESS

- [R1] Scaffolding Human-AI Intelligence via Metacognitive Forcing Functions; with S. Melumad, Z. Bucinca, and M. Groh.
- [R2] Can AI Strengthen Frontline Manager Communication? Evidence from a Field Experiment in Retail; with A. Nyshadham, A. Adhvaryu, and M. Groh.
- [R3] The Human Versatility Advantage over Machines in Ethical Dilemmas; with A. Creo, B. Uzzi, and M. Groh.

PRESENTATIONS AND TALKS

Invited Talks

1. AoM Symposium: Augmenting Human Potential with Generative AI: Opportunities and Risks. *Organizer & Speaker, AoM 2026*
2. Building custom tools for studying social behavior. *Invited speaker, PDW on GenAI as a Window into Organizations: A Window into Cognitive and Social Behavior, AoM 2026*
3. Cultivating Human Empathy with AI Role-playing Partners. *Mind and Machine Alignment Summit, Ohio State University, July 2026*
4. Cultivating Human Empathy with AI Role-playing Partners. *AI and Social Research: Empathic AI, Metascience, and Methodology, Penn State University, June 2026*
5. Enabling Physician-AI Collaboration by Mapping Diagnostic Blind spots & Edges. *MORS Brownbag, May 2026*
6. Practicing with Language Models Cultivates Human Empathic Communication. *Guest Speaker, COG SCI 366: Cognitive Science Research Seminar, Northwestern University, April 2026*

^{1*} indicates equal contribution

7. Cultivating Human Empathy with AI Partners. *Invited Speaker, DySoc Seminar 2025, University of Tennessee, Knoxville*
8. Cultivating Human Empathy with AI Partners. *Guest Speaker, Learn Behavioral Leadership Symposium, Chicago, 2025*
9. Large language models can provide expert-aligned judgments of empathic communication. *Lightning Talk, Northwestern Institute on Complex Systems Seminar, 2025*
10. What Decision-Makers Should Know About Generative AI. *JTIP Symposium 2025: Innovation, AI, and Access, Northwestern School of Law*
11. AI-assisted Decision Making. *UC Noyce Initiative Symposium, 2022*
12. Understanding Human-AI Teams. *Noyce Symposium, 2022*
13. Member, Panel on Human-AI Relationships. *Noyce Symposium, 2022*

Conference Talks

1. Mapping the Idioms of Empathy: A Data-Driven Taxonomy of Empathic Communication Using k-Sparse Autoencoders. *12th International Conference on Computational Social Science (IC2S2), Burlington, VT, July 2026*
2. Practicing with Language Models Cultivates Human Empathic Communication. *12th International Conference on Computational Social Science (IC2S2), Burlington, VT, July 2026*
3. Cultivating Human Empathy with AI Role-playing Partners. *Parallel Talk, Conference on Digital Experimentation @ MIT, 2025*
4. Seeing Things Differently: The Role of Differing Perspectives in Advice-Taking. *Oral Talk, 47th Annual Meeting of the Cognitive Science Society, San Francisco, August 2025*
5. Metacognitive Bandits: When Do Humans Seek AI Assistance? *Social Intelligence in Humans and Robots Workshop, ICRA 2021*
6. Explaining Algorithm Aversion with Metacognitive Bandits. *Spotlight talk, Workshop on Human-AI Collaboration in Sequential Decision-Making, ICML 2021*
7. Explaining Algorithm Aversion with Metacognitive Bandits. *Oral Talk, 43rd Annual Meeting of the Cognitive Science Society, August 2021*

Conference Posters

1. Practicing with Language Models Cultivates Human Empathic Communication. *Symposium on Communication & Intelligence, University of Chicago, Chicago, IL, May 2026*
2. Practicing with Language Models Cultivates Human Empathic Communication. *Wharton AI & the Future of Work Conference, May 2026*
3. When Large Language Models are Reliable for Judging Empathic Communication. *International Conference on Computational Social Science (IC2S2), Norrkoping, Sweden, July 2025*
4. Evaluating Elements of Empathic Communication with Experts, Crowds, and Large Language Model. *International Conference on Computational Social Science (IC2S2), Norrkoping, Sweden, July 2025*
5. Evaluating Human Perception of AI-Generated Images Across Pose Complexities. *International Conference on Computational Social Science (IC2S2), Philadelphia, PA, July 2024*
6. An Empirical Investigation of Reliance on AI-Assistance in a Noisy-Image Classification Task. *Workshop on Trust and Reliance in AI-Human Teams, CHI, April 2023*

AWARDS AND FELLOWSHIPS

UCI Associate Dean's Fellowship 2023, UCI	<i>Total Award: \$7000</i>
Noyce Initiative in AI, Law, and Society Fellowship 2021-2023	<i>Total Award: \$66,000</i>
Graduate Dean's Recruitment Fellowship 2018, UCI	<i>Total Award: \$5000</i>
Recruitment Fellowship Award, 2018, Department of Cognitive Science, UCI	<i>Total Award: \$5000</i>

INDUSTRY EXPERIENCE

Quantitative Human-Computer Interaction Intern, Motional

Apr'23 - Dec'23

Human Assistance in Self-Driving Robotaxis: Investigated Motional's remote vehicle assistance capability in assisting their robotaxi. [P4]

Human Behavior Modeling Intern, Honda Research Institute

Jun'22 - Sept'22

Trust in Autonomous Vehicles: Created a quantitative model to capture the effect of an autonomous vehicle's driving style on users' trust and risk perception. Identified scenarios perceived as risky across two mobility types [8]

Statistical Consultant, Google Research, India

Jan'20 - Jan'21

AI for Social Good: Helped design data collection and analysis methods to evaluate an ML solution's efficacy in reducing drop-off from an Indian maternal health program.

Reliability in People's Inhibitory Control: Proposed and tested RNN models to assess people's cognitive control task traits are reliable. [12]

TEACHING EXPERIENCE

Teaching Assistant, UC Irvine: Probability and Statistics in Psychology I, II, III *Fall 2019 - Spring 2020*

Teaching Assistant, UC Irvine: Creativity *Spring 2019*

Lead Teaching Assistant, UC Irvine: Introduction to Human Memory *Winter 2019*

Teaching Assistant, UC Irvine: Introduction to Psychology *Fall 2018*

MEMBERSHIPS AND SERVICE

Academy of Management Member & Symposium Organizer *2026*

Judge, CoDEX 2026 Poster Session, Northwestern University *2026*

Department Colloquium Coordination Committee Member *2021 - 2022*

Reviewer (Academy of Management, Transactions on Interactive Intelligent Systems, Trends in Cognitive Sciences, Cognitive Science, Human-Robot Interaction, CHI, Applied Psychology: Health and Well-Being, Judgment and Decision-making) *2020 - Present*

REFERENCES

Matthew Groh

Assistant professor at Northwestern University in the Management and Organizations department at the Kellogg School of Management and by courtesy in the Computer Science department at the McCormick School of Engineering.

Email: matthew.groh@kellogg.northwestern.edu

Mark Steyvers

Professor and Chair of Cognitive Sciences at the University of California, Irvine, with an affiliated appointment in Computer Science.

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Shiri Melumad

Associate Professor of Marketing at the Wharton School of the University of Pennsylvania.

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